

Industrial Safety

Chemical Safety



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Contents

1.	INTRODUCTION.....	3
2.	WHAT IS CHEMICAL HANDLING?	4
3.	MAJOR CHEMICAL HAZARDS.....	5
4.	EFFECT ON THE HUMAN BODY	6
5.	RISKS & IMPACTS	7
6.	CHEMICAL ACCIDENTS.....	8
7.	SAFE PRACTICES	9
7.1	SAFETY DO'S & DON'TS IN HANDLING	9
7.2	INVENTORY & CLASSIFICATION	10
7.3	MATERIAL SAFETY DATA SHEET (MSDS)	11
7.4	LABELLING & STORAGE	12
7.5	FIRST AID FOR CHEMICAL EXPOSURE	12
8.	PPE.....	13
9.	PPE FOR HAZARDOUS CHEMICALS	14
10.	SUMMARY	15

1. Introduction

Chemicals are used in manufacturing for various industrial processes, that include:

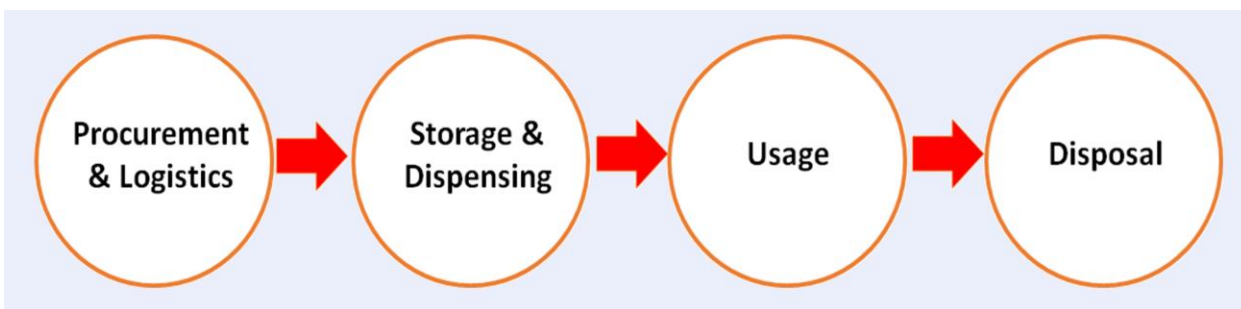
- ▶ **Raw Material:** Many chemicals such as polymers, metals, and solvents, which are transformed into finished products.
- ▶ **Catalysts & Additives:** Chemicals used as catalysts to reactions, or additives to improve product efficiency, quality, strength, durability and performance.
- ▶ **Cleaning & Surface Treatment:** Chemicals used for cleaning, degreasing, and surface treatment of products.
- ▶ **Quality Control:** Chemical analysis of materials to assess raw materials and products properties against quality standards.
- ▶ **Water Treatment:** Chemicals used in water purification and wastewater treatment for reuse.
- ▶ **Preservation & Packaging:** Chemicals are used for preservation, to extend shelf life of products and protect them from environmental factors.
- ▶ **Waste Disposal:** Chemicals are used to deactivate and dispose harmful chemical waste and prevent harming the environment.
- ▶ **Product Research:** Ongoing research and innovation in chemical applications.
- ▶ **Most Machinery, Tools & Appliances** are powered by Electricity. Operating such equipment requires qualified personnel following established safety norms.



2. What is Chemical Handling?

You are handling chemicals if you are in any way involved in its

- ▶ Procurement & Transportation
- ▶ Storing it in Warehouses and Dispensing them to Production
- ▶ Production using Chemicals
- ▶ Disposal of Waste Chemicals



Handling Chemicals – Examples



Chemical storage
in warehouse



Dispensing
chemicals using
spill platform



Chemical container
labelling

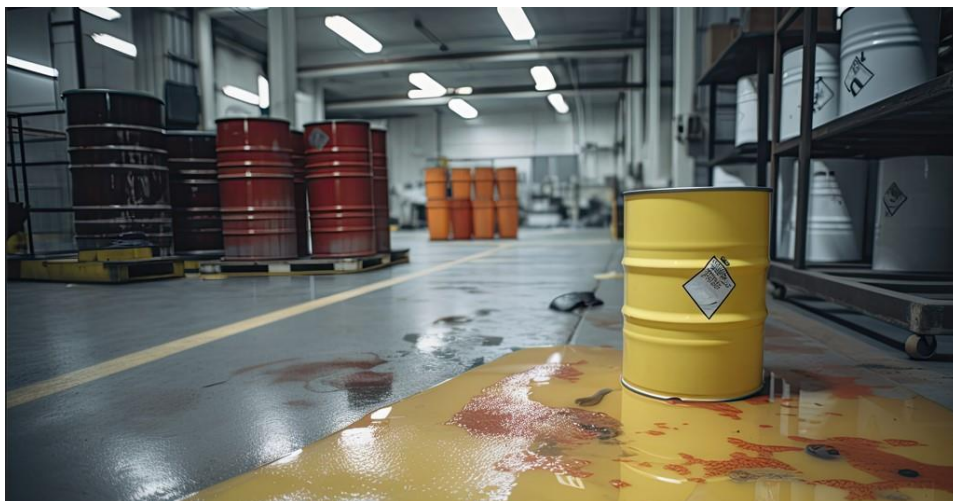


Handling
hazardous
chemicals

3. Major Chemical Hazards

Some of the major hazards from chemicals include:

- ▶ **Faulty equipment:** leading to equipment failure and leaks
- ▶ **Transportation accidents:** causing leaks and spills
- ▶ **Improper storage conditions:** leading to ignition and reactions
- ▶ **Improper handling:** due to inadequate training and awareness
- ▶ **Inadequate control of ignition sources:** near storage areas
- ▶ **Exposure: or contact** to hazardous chemicals
- ▶ **Inadequate PPE:**
- ▶ **Inadequate Emergency plans:** & response leading to escalations

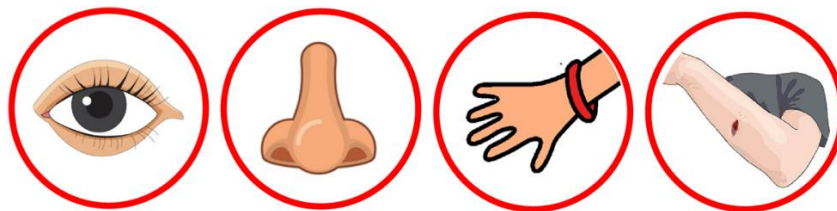


4. Effect on the Human Body

Most chemicals are corrosive and flammable. They can cause adverse effects on the body when in direct contact.



Chemicals, can enter the body through the Eyes, Nose, Skin and through wounds and cause ill effects to the body.



5. Risks & Impacts

Health Risks from Chemicals could be:

- ▶ Be an irritant and cause respiratory problems
- ▶ Cause wounds to skin, eyes, lungs
- ▶ Cause poisoning, chronic health effects & cancer

Other adverse impacts to the environment could be:

- ▶ Contaminate the Environment
- ▶ Cause health effects to nearby communities
- ▶ Harmful byproducts can cause heat, fire and explosions



6. Chemical Accidents

Here are some of potential accidents that can occur in the manufacturing industry:

- ▶ **Chemical Spills and Leaks:** due to accidental release of chemicals during transportation, storage, processing or handling.
- ▶ **Fire and Explosions:** due to accidental ignition.
- ▶ **Toxic Gas Releases:** Accidental releases of toxic gases from manufacturing processes.
- ▶ **Unintended Chemical Reactions:** leading to fire and explosions.
- ▶ **Exposure to Hazardous Substances:** Exposure of workers to hazardous substances through skin contact, inhalation, or ingestion can lead to chronic health issues.



7. Safe Practices

Here are some Safe Practices in Handling Chemicals

- ▶ Safety Do's & Don'ts in Handling
- ▶ Inventory & Classification
- ▶ MSDS
- ▶ Labelling & Storage



7.1 Safety Do's & Don'ts in Handling

Do's:

- ▶ Only Trained People should handle Hazardous chemicals
- ▶ Storage locations should be cool, dry place away from heat and sunlight.
- ▶ Every chemical container should be labeled
- ▶ Container should be of compatible to chemicals it holds.
- ▶ An inventory list of all chemicals on the site.
- ▶ MSDS available for all chemicals on site should be maintained.
- ▶ Transfer flammable chemicals using non-electrostatic tools.
- ▶ Always wash hands after using any chemical.
- ▶ Always wear prescribed PPE

Don'ts

- ▶ Don't mix or combine hazardous chemicals.
- ▶ No eating, drinking or smoking near chemicals.
- ▶ Never pour Chemicals down the drains.

7.2 Inventory & Classification

An updated Inventory of all Hazardous Chemicals available at site should always be maintained.

Chemical Inventory Table					
Area in Factory	Name	Amount on hand	Scale of use g/kg/ton ml/l/m ³	Safety Information Hazard Band	Notes about handling, use, storage
Chemicals in the stoking area					
Chemicals being transferred and handle					
Chemicals involved in the Production Process					
Chemicals stored in places other than stoking area					
Chemicals in the solid waste					
Chemicals in waste water					
Chemicals in the air (emissions)					

Inventory							
Cabinet 1, Room 205, Building 110				Add Inventory Print Inventory			
Chemical Name	Date Received	Amount Stored	Storage Code	CAS#	Formula	MSDS	Delete
Ammonium Chloride	08/15/2009	100 g	Green	12125-02-9	NH ₄ Cl		
Bromothymol Blue, sodium salt	12/13/2008	125 g	Green	34722-90-2	C ₂₇ H ₂₇ Br ₂ O ₅ Na		
Luminol powder	11/02/2009	1 g	Green	521-31-3	C ₈ H ₇ N ₃ O ₂		
Potassium Chloride 1.0M Aq soln	03/07/2010	50 ml	Green		Mixture		
Sodium Chloride	08/22/2009	425 g	Green	7647-14-5	NaCl ₂		
Submit Changes Reset							

7.3 Material Safety Data Sheet (MSDS)

- ▶ A Material Safety Data Sheet (MSDS) is a document that contains information on the potential hazards and how to work safely with the chemical product.
- ▶ The MSDS should be maintained against stock of all stocked Chemicals, especially of Hazardous Chemicals.



7.4 Labelling & Storage

- Each container must be labelled with the identity of the hazardous chemicals along with appropriate hazard warnings.
- Whenever a chemical is transferred to an unlabeled container, a new label must be created and securely attached to the container.

Category/ Category						
Flammable		X	X	X	X	X Very dangerous combination, never store together
Oxidizing	X		X	X	X	X Dangerous combination, avoid storing together
Toxic	X	X		X	X	X Only store together if compounds do not react with each other
Corrosive	X	X	X		X	
Dangerous for the environment	X	X	X	X		

Physical Hazards				
				
Explosives	Flammable Liquids	Oxidizing Liquids	Compressed Gases	Corrosive to Metals
Health Hazards				Env. Hazards
				
Acute Toxicity	Skin Corrosion	Skin Irritation	CMR's, STOT's, Aspiration Hazard	Hazardous to the Aquatic Environment

7.5 First Aid for Chemical Exposure

Emergency showers and eyewash stations provide on-the-spot decontamination. They allow workers to flush away hazardous substances on their body that can cause injury. On contact with any hazardous chemical, immediately:

- Stop the source
- Clear the lungs
- Flush the eyes
- Clean the skin



8. PPE

PPE that are required to be worn when working with Chemicals include:

- ▶ Eye glasses
- ▶ Disposable N95 mask
- ▶ Chemical resistant Hand gloves
- ▶ Chemical resistant apron
- ▶ Boots



9. PPE for Hazardous Chemicals

When working with any hazardous materials, always consult the Material's MSDS for recommended PPE. These should include:

- ▶ Full-face respirator with chemical-resistant lenses
- ▶ Chemical-resistant headgear or hoods
- ▶ Chemical-resistant gloves
- ▶ Chemical-resistant coveralls
- ▶ Chemical-resistant boots with boot covers



10. Summary

- ▶ Chemicals are necessary in the production of most products.
- ▶ Most Chemicals cause harm to people and the environment if not handled properly.
- ▶ Some Hazardous Chemicals need to be handled by specialized techniques to prevent harm and damage to property.
- ▶ The full-cycle of Chemical Handling needs to be professionally and safely managed.

